

ENGINEERING MATHEMATICS-II

UNIT 1: INTEGRAL CALCULUS

Class 11: MASS AND CENTER OF MASS

DEFINITION OF MASS OF PLANAR LAMINA

Double integrals can be used to find the mass of a lamina of variable density, where the density at (x, y) is given by the **density function** ρ .

If ρ is a continuous density function on the lamina corresponding to a plane region R , then the mass m of the lamina is given by

$$m = \iint_R \rho(x, y) dA.$$

Variable density

PROBLEM ON MASS USING DOUBLE INTEGRAL

Example 1: Find the mass of a lamina with density function $\rho(x, y) = 6x$ covering the triangle D bounded by the x -axis, the line $y = x$, and the line $y = 2 - x$.

$$\begin{aligned} M &= \iint_D \rho(x, y) dA \\ &= \int_0^1 \int_y^{2-y} 6x \, dx \, dy \\ &= \int_0^1 3[(2-y)^2 - y^2] dy \\ &= -[(2-y)^3 + y^3]_0^1 = 6. \end{aligned}$$

CENTER OF MASS

The center of mass is also known as the center of gravity, if the object has uniform density, the center of mass is the geometric center of the object, which is called the centroid. [Figure 5.64](#) shows a point P as the center of mass of a lamina. The lamina is perfectly balanced about its center of mass.

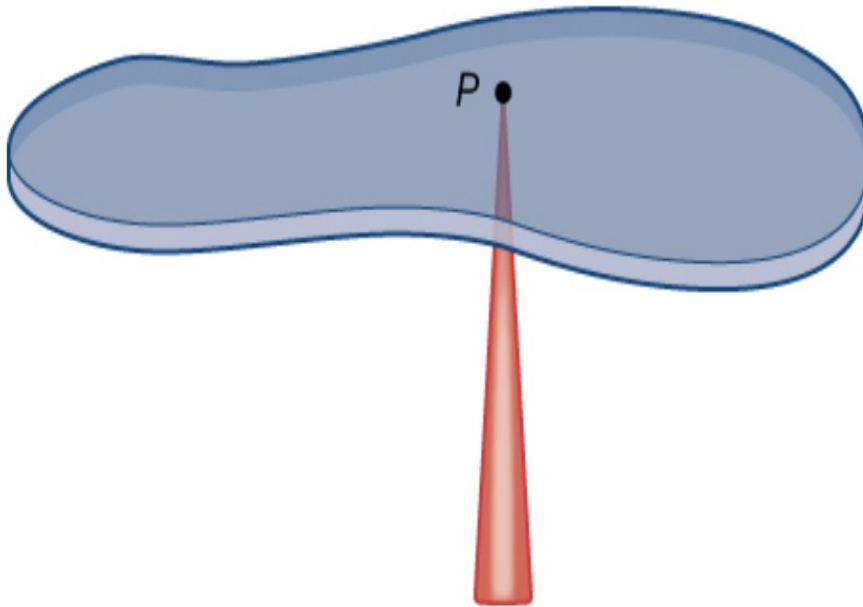


Figure 5.64 A lamina is perfectly balanced on a spindle if the lamina's center of mass sits on the spindle.

CENTER OF MASS

Let ρ be a continuous density function on the planar lamina R . The **moments of mass** with respect to the x - and y -axes are

$$M_x = \iint_R y\rho(x, y) dA \quad \text{and} \quad M_y = \iint_R x\rho(x, y) dA.$$

If m is the mass of the lamina, then the **center of mass** is

$$(\bar{x}, \bar{y}) = \left(\frac{M_y}{m}, \frac{M_x}{m} \right).$$

If R represents a simple plane region rather than a lamina, the point $(\bar{x}, \bar{y})$ is called the **centroid** of the region.

PROBLEM ON CENTER OF MASS USING DOUBLE INTEGRAL

Find the center of mass of a square lamina, represented by the unit square with lower lefthand corner at the origin with variable density $\delta(x, y) = (x + y + 2)\text{g/cm}^2$.

$$M = \iint_R (x + y + 2) \, dA = \int_0^1 \int_0^1 (x + y + 2) \, dx \, dy = 3g.$$

$$M_x = \iint_R y(x + y + 2) \, dA = \int_0^1 \int_0^1 y(x + y + 2) \, dx \, dy = \frac{19}{12}.$$

$$M_y = \iint_R x(x + y + 2) \, dA = \int_0^1 \int_0^1 x(x + y + 2) \, dx \, dy = \frac{19}{12}.$$

$$\text{Thus the center of mass is } (\bar{x}, \bar{y}) = \left(\frac{M_y}{M}, \frac{M_x}{M} \right) = \left(\frac{19}{36}, \frac{19}{36} \right).$$

MASS OF A SOLID USING TRIPLE INTEGRAL AND MOMENTS ABOUT THE PLANES.

If we have a solid object Q with a density function at any point (x, y, z) in space, then its mass is

$$m = \iiint_Q \rho(x, y, z) dV.$$

Its moments about the xy -plane, the xz -plane and the yz -plane

$$M_{xy} = \iiint_Q z \rho(x, y, z) dV, \quad M_{xz} = \iiint_Q y \rho(x, y, z) dV, \\ M_{yz} = \iiint_Q x \rho(x, y, z) dV.$$

If the center of mass of the object is the point $(\bar{x}, \bar{y}, \bar{z})$

$$\bar{x} = \frac{M_{yz}}{m}, \quad \bar{y} = \frac{M_{xz}}{m}, \quad \bar{z} = \frac{M_{xy}}{m}.$$

EXAMPLE 1: MASS

An object occupies the space inside both the cylinder $x^2 + y^2 = 1$ and the sphere $x^2 + y^2 + z^2 = 4$, and has density x^2 at (x, y, z) . Find the total mass.

We set this up in cylindrical coordinates, recalling that $x = r \cos \theta$:

$$\begin{aligned} \int_0^{2\pi} \int_0^1 \int_{-\sqrt{4-r^2}}^{\sqrt{4-r^2}} r^3 \cos^2(\theta) \, dz \, dr \, d\theta &= \int_0^{2\pi} \int_0^1 2\sqrt{4-r^2} \, r^3 \cos^2(\theta) \, dr \, d\theta \\ &= \int_0^{2\pi} \left(\frac{128}{15} - \frac{22}{5} \sqrt{3} \right) \cos^2(\theta) \, d\theta \\ &= \left(\frac{128}{15} - \frac{22}{5} \sqrt{3} \right) \pi \end{aligned}$$

EXAMPLE 2: CENTER OF MASS

Suppose Q is a solid region bounded by the plane $x + 2y + 3z = 6$ and the coordinate planes with density $\rho(x, y, z) = x^2yz$. Find the center of mass using decimal approximation.

$$M_{xy} = \iiint_Q z\rho(x, y, z) dV = \int_{x=0}^6 \int_{y=0}^{1/2(6-x)} \int_{z=0}^{1/3(6-x-2y)} x^2yz^2 dz dy dx = \frac{54}{35} \approx 1.543,$$

$$M_{xz} = \iiint_Q y\rho(x, y, z) dV = \int_{x=0}^6 \int_{y=0}^{1/2(6-x)} \int_{z=0}^{1/3(6-x-2y)} x^2y^2z dz dy dx = \frac{81}{35} \approx 2.314,$$

$$M_{yz} = \iiint_Q x\rho(x, y, z) dV = \int_{x=0}^6 \int_{y=0}^{1/2(6-x)} \int_{z=0}^{1/3(6-x-2y)} x^3yz dz dy dx = \frac{243}{35} \approx 6.943.$$

EXAMPLE 2: CENTER OF MASS

Hence the center of mass is

$$\bar{x} = \frac{M_{yz}}{m}, \bar{y} = \frac{M_{xz}}{m}, \bar{z} = \frac{M_{xy}}{m},$$

$$\bar{x} = \frac{M_{yz}}{m} = \frac{243/35}{108/35} = \frac{243}{108} = 2.25,$$

$$\bar{y} = \frac{M_{xz}}{m} = \frac{81/35}{108/35} = \frac{81}{108} = 0.75,$$

$$\bar{z} = \frac{M_{xy}}{m} = \frac{54/35}{108/35} = \frac{54}{108} = 0.5.$$

The center of mass for the tetrahedron Q is the point $(2.25, 0.75, 0.5)$.



THANK YOU
